

Orthospace: An Operator-Theoretic Extension of Shadow Math and UCSIL

Abstract

We formalize *Orthospace* as a new structural component in the Shadow Math / UCSIL (Unified Calculus of Shadows, Information, and Learning) framework. Starting from a pair of Hilbert spaces for identity and shadows, and a self-projection operator defined as a composition of shadowing and reconstruction, we introduce a canonical decomposition of the self-map into symmetric (drift/learning) and skew-adjoint (rotational/self-structure) parts. The skew-adjoint component generates a unitary “self-rotation” flow on identity space, and we define Orthospace as the closed image of this orthogenerator.

We develop the spectral theory, dynamical properties, scalar metrics (orthointensity, orthovolume, orthogradients), discrete-time analogues, and data-driven estimation procedures, and we extend the construction to multi-agent meshes. We further relate Orthospace to information geometry, provide a category-theoretic perspective on shadow systems, give nonlinear manifold extensions, and work out concrete examples. We also introduce canonical scalar summary statistics, Cayley-transform-based discrete orthoflow, and mesh-level orthocoherence measures that are immediately usable for large-scale identity systems. This document is self-contained and can be used as a chapter-level insert for a broader Shadow Math / UCSIL monograph.

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1 Base Setup: Identity, Shadow, and Self-Map

We begin by formalizing the identity and shadow layers as Hilbert spaces and introducing the self-projection map as a composition of a shadowing operator and a reconstruction operator.

1.1 Identity and shadow spaces

Assumption 1.1 (Identity and Shadow Spaces). *There exist separable Hilbert spaces*

$$\mathcal{I}, \mathcal{S}$$

over $\mathbb{C}$, equipped with inner products

$$\langle \cdot | \cdot \rangle_{\mathcal{I}} \quad \text{and} \quad \langle \cdot | \cdot \rangle_{\mathcal{S}},$$

representing, respectively, the identity space (internal state of a system) and the shadow space (externally realized/observable states).

Intuitively, $\mathcal{I}$ encodes high-dimensional internal “identity” configurations, while $\mathcal{S}$ encodes lower-dimensional observable projections.

1.2 Shadowing and reconstruction operators

Assumption 1.2 (Projection and Model Operators). *There exist bounded linear operators*

$$\Pi : \mathcal{I} \rightarrow \mathcal{S}, \quad M : \mathcal{S} \rightarrow \mathcal{I}$$

where

- Π is the shadowing operator, mapping identity states to observable shadows;
- M is the model/reconstruction operator, mapping observed shadows into the system’s internal representation of those shadows in $\mathcal{I}$.

These two maps are the functional-analytic realization of the shadow process:

$$\text{Identity} \xrightarrow{\Pi} \text{Shadow} \xrightarrow{M} \text{Internal Model}.$$

1.3 Self-projection map

Definition 1.3 (Self-Projection Map). *The self-projection map (or self-map) induced by “project then reconstruct” is defined as*

$$\Theta := M \circ \Pi : \mathcal{I} \rightarrow \mathcal{I}.$$

For $x \in \mathcal{I}$, the interpretation is:

- (i) start from internal identity x ,
- (ii) produce its shadow $\Pi x \in \mathcal{S}$,
- (iii) reconstruct that shadow via M , producing $M(\Pi x) \in \mathcal{I}$,
- (iv) the net effect on identity is Θx .

We assume Θ is bounded; this holds automatically if both M and Π are bounded.

2 Adjoint and Orthogonal Decomposition of the Self-Map

The self-map Θ admits a canonical decomposition into symmetric and skew-adjoint parts with respect to the inner product on $\mathcal{I}$.

2.1 Adjoint of the self-map

Since Θ is a bounded linear operator on a Hilbert space $\mathcal{I}$, there is a unique adjoint operator

$$\Theta^* : \mathcal{I} \rightarrow \mathcal{I}$$

such that

$$\langle \Theta x | y \rangle_{\mathcal{I}} = \langle x | \Theta^* y \rangle_{\mathcal{I}} \quad \text{for all } x, y \in \mathcal{I}.$$

2.2 Symmetric and skew-adjoint components

Definition 2.1 (Symmetric and Skew-Adjoint Parts). *Define operators*

$$\Sigma := \frac{1}{2}(\Theta + \Theta^*), \quad \Omega := \frac{1}{2}(\Theta - \Theta^*).$$

Proposition 2.2. *The operators Σ and Ω satisfy:*

- (i) $\Sigma^* = \Sigma$ (self-adjoint),
- (ii) $\Omega^* = -\Omega$ (skew-adjoint),
- (iii) $\Theta = \Sigma + \Omega$.

Proof. Standard properties of adjoint and linearity:

$$\Sigma^* = \left(\frac{1}{2}(\Theta + \Theta^*) \right)^* = \frac{1}{2}(\Theta^* + \Theta) = \Sigma,$$

and

$$\Omega^* = \left(\frac{1}{2}(\Theta - \Theta^*) \right)^* = \frac{1}{2}(\Theta^* - \Theta) = -\Omega.$$

Clearly $\Theta = \Sigma + \Omega$ by construction. □

Remark 2.3 (Interpretation). • Σ represents a “symmetric” effect on identity: drift, reshaping, alignment, and diffusion-like behaviour.

- Ω represents a purely rotational effect on identity: a norm-preserving reorientation of internal state.

We refer to Ω as the orthogenerator, as it generates the orthorotational dynamics central to Orthospace.

3 Orthospace as the Image of the Orthogenerator

We now define Orthospace as the closed image of the skew-adjoint component Ω .

Definition 3.1 (Orthospace). *Given $\Theta = M \circ \Pi$ and its decomposition $\Theta = \Sigma + \Omega$ with Σ self-adjoint and Ω skew-adjoint, the Orthospace is*

$$\mathcal{O} := \overline{\text{Im}(\Omega)} \subseteq \mathcal{I},$$

the closure of the image of Ω in $\mathcal{I}$.

Remark 3.2. $\mathcal{O}$ consists of all directions in identity space that can be reached through pure antisymmetric self-rotation. By construction,

$$x \in \mathcal{O} \iff x \text{ can be approximated by vectors of the form } \Omega y, \ y \in \mathcal{I}.$$

We also define the subspace on which self-rotation vanishes:

Definition 3.3 (Fixed Subspace). *The fixed subspace of orthorotation is*

$$\mathcal{F} := \ker(\Omega) = \{x \in \mathcal{I} : \Omega x = 0\}.$$

On $\mathcal{F}$, self-reflection via Θ has no rotational component; only the symmetric part Σ acts. On $\mathcal{O}$, the skew-adjoint component generates a “spin” effect.

4 Orthoflow: Dynamics Generated by the Orthogenerator

Because Ω is skew-adjoint, it generates a unitary flow on $\mathcal{I}$.

4.1 Unitary group generated by Ω

Theorem 4.1 (Stone’s Theorem (Specialized)). *If Ω is skew-adjoint on the Hilbert space $\mathcal{I}$, then there exists a unique one-parameter strongly continuous unitary group $\{U(t)\}_{t \in \mathbb{R}}$ such that*

$$U(t) = e^{t\Omega}, \quad t \in \mathbb{R},$$

and for all $x \in \mathcal{I}$, the map $t \mapsto U(t)x$ is differentiable and satisfies

$$\frac{d}{dt}U(t)x = \Omega U(t)x.$$

Remark 4.2. Since Θ is assumed bounded, Θ^* exists and is bounded, so Ω is bounded and skew-adjoint. In this setting, Stone’s theorem applies directly, and the resulting unitary group $e^{t\Omega}$ is defined on the whole of $\mathcal{I}$.

4.2 Orthoflow dynamics

Definition 4.3 (Orthoflow). *The orthoflow on $\mathcal{I}$ generated by Ω is the differential equation*

$$\frac{d}{dt}x(t) = \Omega x(t), \quad x(0) = x_0 \in \mathcal{I}.$$

Its unique solution is

$$x(t) = U(t)x_0 = e^{t\Omega}x_0.$$

Proposition 4.4 (Properties of Orthoflow). *For $x(t) = e^{t\Omega}x_0$ we have:*

- (i) (Norm preservation)
$$\|x(t)\| = \|x_0\| \quad \text{for all } t \in \mathbb{R}.$$
- (ii) (Invariance of $\mathcal{O}$) If $x_0 \in \mathcal{O}$, then $x(t) \in \mathcal{O}$ for all t .
- (iii) (Pure self-rotation) Changes in $x(t)$ are internal reorientations induced by self-reflection, with no external input.

Proof. Unitary operators preserve norms, so (i) holds. Since $\mathcal{O}$ is the closed image of Ω and $U(t)$ commutes appropriately with Ω via the functional calculus, $\mathcal{O}$ is invariant, giving (ii). (iii) is an interpretation of these properties. $\square$

5 Full Self-Dynamics: Drift and Rotation

The complete self-update dynamics of identity induced by the self-map Θ can be written as a combination of drift (symmetric) and rotation (skew-adjoint) components.

5.1 Continuous-time self-dynamics

Consider the evolution equation

$$\frac{d}{dt}x(t) = \Theta x(t) = \Sigma x(t) + \Omega x(t), \quad x(0) = x_0 \in \mathcal{I}. \quad (1)$$

Remark 5.1. • Σ can be interpreted as gradient-like or diffusion-like, potentially associated with learning, convergence, or alignment.

- Ω is responsible for norm-preserving rotations orthogonal to these gradient directions, shaping the internal structure and “narrative” of identity.

5.2 Energy interpretation

Define the identity energy

$$E(x) := \frac{1}{2}\|x\|^2.$$

Proposition 5.2 (Orthoflow Is Energy-Preserving). *For any $x \in \mathcal{I}$,*

$$\langle x | \Omega x \rangle_{\mathcal{I}} = 0,$$

so orthoflow does not change the energy $E(x)$.

Proof. Since $\Omega^* = -\Omega$,

$$\langle x | \Omega x \rangle = \langle \Omega^* x | x \rangle = \langle -\Omega x | x \rangle = -\overline{\langle x | \Omega x \rangle}.$$

Hence $\langle x | \Omega x \rangle$ is purely imaginary, so its real part (which determines the time-derivative of $\|x\|^2$) vanishes. $\square$

For the full dynamics (1), differentiating $E(x(t))$ yields

$$\frac{d}{dt}E(x(t)) = \langle x(t) | \Sigma x(t) \rangle + \langle x(t) | \Omega x(t) \rangle.$$

The second term has zero real part, so all change in E is due to Σ :

$$\frac{d}{dt}E(x(t)) = \text{Re} \langle x(t) | \Sigma x(t) \rangle.$$

Thus:

- Σ controls norms (growth/shrinkage),
- Ω controls orientation (self-rotation).

6 Orthometric Quantities: Orthointensity, Orthovolume, Orthogradients

We now define scalar functionals that capture the magnitude and distribution of orthorotational behaviour.

6.1 Orthointensity

Definition 6.1 (Orthointensity). *The orthointensity associated with Ω at state $x \in \mathcal{I}$ is*

$$\mathcal{I}_\Omega(x) := \|\Omega x\|^2 \geq 0.$$

This measures how strongly the identity state x is currently undergoing self-rotation.

6.2 Population orthovolume

Let μ be a probability measure on $\mathcal{I}$ (e.g., representing the distribution of identity states across time or across agents). We define:

Definition 6.2 (Population Orthovolume). *The orthovolume of μ is*

$$\mathcal{V}_\Omega(\mu) := \int_{\mathcal{I}} \|\Omega x\|^2 d\mu(x).$$

This can be viewed as a total “orthokinetic energy” of a population of identities.

6.3 Orthogradient of a functional

Let $F : \mathcal{I} \rightarrow \mathbb{R}$ be Fréchet differentiable with gradient $\nabla F(x) \in \mathcal{I}$. We define:

Definition 6.3 (Orthogradient). *The orthogradient of F at x is*

$$\nabla_\Omega F(x) := \Omega(\nabla F(x)).$$

Consider a trajectory $x(t)$ following orthoflow: $x'(t) = \Omega x(t)$. Then

$$\left. \frac{d}{dt} F(x(t)) \right|_{t=0} = \langle \nabla F(x) | \Omega x \rangle.$$

This is the directional derivative of F along pure self-rotation.

7 Axiomatization for Shadow Math / UCSIL

We summarize the core structure as a set of axioms, suitable for inclusion in a Shadow Math / UCSIL foundational chapter.

Assumption 7.1 (O1: Identity and Shadow Spaces). *There exist separable Hilbert spaces $\mathcal{I}$ and $\mathcal{S}$, representing identity and shadow spaces, respectively.*

Assumption 7.2 (O2: Projection & Model). *There exist bounded linear operators*

$$\Pi : \mathcal{I} \rightarrow \mathcal{S}, \quad M : \mathcal{S} \rightarrow \mathcal{I},$$

representing shadowing and reconstruction.

Assumption 7.3 (O3: Self-Map). *The self-projection map is*

$$\Theta := M \circ \Pi : \mathcal{I} \rightarrow \mathcal{I}.$$

Assumption 7.4 (O4: Orthodecomposition). *The operator Θ admits the decomposition*

$$\Theta = \Sigma + \Omega,$$

with

$$\Sigma := \frac{1}{2}(\Theta + \Theta^*), \quad \Omega := \frac{1}{2}(\Theta - \Theta^*)$$

such that Σ is self-adjoint and Ω is skew-adjoint.

Assumption 7.5 (O5: Orthospace). *The Orthospace is defined as*

$$\mathcal{O} := \overline{\text{Im}(\Omega)},$$

and we impose

$$\mathcal{I} = \ker(\Omega) \oplus \mathcal{O}$$

either by assuming Ω is normal or as a structural axiom on the system.

These five axioms yield a clean, publishable mathematical object and extend the Shadow Math / UCSIL framework with a rigorous orthodimensional component.

8 Finite-Dimensional Example in $\mathbb{R}^2$

We now provide a concrete example in $\mathbb{R}^2$ to illustrate the constructions.

Let

$$\mathcal{I} = \mathbb{R}^2, \quad \mathcal{S} = \mathbb{R}.$$

Define:

- Projection: $\Pi(x, y) = x$ (only first coordinate is observable).
- Model: $M(s) = (s, \alpha s)$ for some fixed $\alpha \in \mathbb{R}$.

Then the self-map $\Theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is

$$\Theta(x, y) = M(\Pi(x, y)) = M(x) = (x, \alpha x),$$

so in matrix form:

$$\Theta = \begin{pmatrix} 1 & 0 \\ \alpha & 0 \end{pmatrix}.$$

Under the standard Euclidean inner product, the adjoint is the transpose:

$$\Theta^* = \Theta^\top = \begin{pmatrix} 1 & \alpha \\ 0 & 0 \end{pmatrix}.$$

Compute:

$$\begin{aligned} \Sigma &= \frac{1}{2}(\Theta + \Theta^*) = \frac{1}{2} \begin{pmatrix} 1+1 & 0+\alpha \\ \alpha+0 & 0+0 \end{pmatrix} = \begin{pmatrix} 1 & \alpha/2 \\ \alpha/2 & 0 \end{pmatrix}, \\ \Omega &= \frac{1}{2}(\Theta - \Theta^*) = \frac{1}{2} \begin{pmatrix} 1-1 & 0-\alpha \\ \alpha-0 & 0-0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & -\alpha \\ \alpha & 0 \end{pmatrix}. \end{aligned}$$

Thus

$$\Omega = \frac{\alpha}{2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

which is precisely the standard 2×2 rotation generator scaled by $\alpha/2$.

Proposition 8.1. *If $\alpha \neq 0$, then Ω generates rotations in $\mathbb{R}^2$:*

$$e^{t\Omega} = \begin{pmatrix} \cos(\omega t) & -\sin(\omega t) \\ \sin(\omega t) & \cos(\omega t) \end{pmatrix}, \quad \omega = \alpha/2.$$

In this case $\mathcal{O} = \mathbb{R}^2$, and all identity states undergo self-induced rotation from self-projection, with strength proportional to α .

Proof. Standard matrix exponential of a 2×2 skew-symmetric matrix yields the rotation matrix with angular frequency $\omega = \alpha/2$. Since Ω has full rank for $\alpha \neq 0$, its image is all of $\mathbb{R}^2$, so $\mathcal{O} = \mathbb{R}^2$. $\square$

This simple example concretely illustrates the notion that self-reflection (Θ via $M \circ \Pi$) can generate an internal rotational dynamics on the identity space.

9 Spectral Structure of the Orthogenerator

We next study the spectral properties of Ω and the induced decomposition of identity into fixed and rotating components.

9.1 Spectrum of a skew-adjoint operator

Proposition 9.1 (Imaginary Spectrum). *Let Ω be skew-adjoint on $\mathcal{I}$. Then its spectrum is contained in the imaginary axis:*

$$\sigma(\Omega) \subseteq i\mathbb{R}.$$

Definition 9.2 (Orthospectrum). *The orthospectrum of the system is the multiset*

$$\text{Spec}_\Omega := \{\lambda \in i\mathbb{R} : \exists x \neq 0, \Omega x = \lambda x\}.$$

Each eigenvalue $\lambda = i\omega$ with $\omega \in \mathbb{R}$ represents a rotational mode with angular frequency $|\omega|$.

9.2 Finite-dimensional block structure

Proposition 9.3 (Canonical Rotation Blocks (Finite Dimension)). *If $\mathcal{I}$ is finite-dimensional and Ω is skew-Hermitian, then there exists an orthonormal basis of $\mathcal{I}$ in which Ω decomposes into a direct sum of 1×1 zero blocks and 2×2 blocks of the form*

$$\begin{pmatrix} 0 & -\omega_j \\ \omega_j & 0 \end{pmatrix}, \quad \omega_j \in \mathbb{R}.$$

Each block corresponds to a 2-dimensional orthoplane on which identity undergoes uniform rotation at angular speed $|\omega_j|$.

Remark 9.4. *In the finite-dimensional case, Orthospace $\mathcal{O}$ is literally the direct sum of these 2D rotational planes; states in $\ker(\Omega)$ lie outside the rotational subspace and remain fixed under orthoflow.*

10 Decomposition of Identity into Inert and Rotating Parts

Under a mild assumption, we can decompose $\mathcal{I}$ as a direct sum of the fixed subspace and the Orthospace.

10.1 Normality and reduced decomposition

Assumption 10.1 (Normality / Structural Axiom). *We assume Ω is normal, i.e.*

$$\Omega\Omega^* = \Omega^*\Omega,$$

or we take the equivalent decomposition as a structural axiom.

Then we use a standard identity for normal operators on Hilbert spaces:

$$\overline{\text{Im}(\Omega)} = (\ker \Omega^*)^\perp = (\ker \Omega)^\perp.$$

Proposition 10.2 (Orthogonal Decomposition). *Under the normality assumption,*

$$\mathcal{I} = \ker(\Omega) \oplus \overline{\text{Im}(\Omega)} = \mathcal{F} \oplus \mathcal{O}.$$

Remark 10.3 (Reduced Orthospace Without Full Normality). *Even when Ω is not globally normal, one can often restrict attention to a reduced subspace $\mathcal{H}_{\text{red}} \subseteq \mathcal{I}$ on which Ω admits an orthogonal decomposition (e.g. via standard reducing-subspace decompositions for bounded operators). In practice, Orthospace analysis is typically performed on this “purely rotational” reduced subspace, where $\ker(\Omega)$ and $\overline{\text{Im}(\Omega)}$ behave orthogonally.*

Thus, any $x \in \mathcal{I}$ (or $x \in \mathcal{H}_{\text{red}}$ in the reduced setting) uniquely decomposes as

$$x = x_{\text{fix}} + x_{\text{ortho}}, \quad x_{\text{fix}} \in \mathcal{F}, \quad x_{\text{ortho}} \in \mathcal{O}.$$

10.2 Dynamics of fixed and orthocomponents

Proposition 10.4 (Dynamics of Fixed and Orthocomponents). *Let $x(t)$ solve the orthoflow $x'(t) = \Omega x(t)$ with $x(0) = x_0$. Writing $x_0 = x_{\text{fix}} + x_{\text{ortho}}$ as above, we have*

$$x(t) = x_{\text{fix}} + e^{t\Omega} x_{\text{ortho}},$$

where x_{fix} is constant in time and $e^{t\Omega} x_{\text{ortho}}$ is a unitary rotation within $\mathcal{O}$.

Remark 10.5. • $\mathcal{F}$: “rigid” identity — deep invariants unaffected by self-rotation.

- $\mathcal{O}$: “live” identity — the part that undergoes continuous orthorotation, shaping narrative, framing, and internal perspective.

11 General Self-Dynamics with External Input

In practice, self-dynamics may include external driving signals and noise.

11.1 Deterministic input

We consider the evolution equation

$$\frac{d}{dt}x(t) = \Theta x(t) + Bu(t),$$

where:

- $x(t) \in \mathcal{I}$ is the identity state,
- $u(t)$ lies in an input space $\mathcal{U}$,
- $B : \mathcal{U} \rightarrow \mathcal{I}$ is a bounded linear operator.

Definition 11.1 (Orthodrive Identity System). *An orthodrive identity system is a triple $(\mathcal{I}, \Theta, B)$ with dynamics*

$$x'(t) = \Theta x(t) + Bu(t),$$

where $\Theta = \Sigma + \Omega$ is the orthodecomposition of the self-map.

11.2 Stochastic extension

If we add a noise term $\eta(t)$ modeling stochastic perturbations, we can write a stochastic differential equation on $\mathcal{I}$:

$$dx_t = \Theta x_t dt + Bu_t dt + dW_t,$$

where W_t is an $\mathcal{I}$ -valued Wiener process (Brownian motion) in the sense of stochastic analysis on Hilbert spaces. This allows modelling orthorotational dynamics under uncertainty.

12 Discrete-Time Version for Logged Systems

Many real systems (e.g., OCTA nodes, P3P mesh nodes) are observed and logged in discrete-time steps.

12.1 Discrete self-map and orthodecomposition

Let $T : \mathcal{I} \rightarrow \mathcal{I}$ be a bounded linear operator representing the update

$$x_{k+1} = Tx_k.$$

Definition 12.1 (Discrete Orthodecomposition). *We define*

$$S := \frac{1}{2}(T + T^*), \quad K := \frac{1}{2}(T - T^*).$$

Then S is self-adjoint, K is skew-adjoint, and $T = S + K$.

The discrete analogue of Orthospace is:

$$\mathcal{O}_{\text{disc}} := \overline{\text{Im}(K)}.$$

Remark 12.2. *In discrete time, K captures the “instantaneous skew” part of T , but the full discrete-time evolution T^k need not be unitary even if K is non-zero. For monitoring logged systems, K is typically interpreted as the discrete orthogenerator, while strict unitary evolution can be recovered via a Cayley transform (see §19.2).*

12.2 Discrete orthometric quantities

In discrete time, we define:

$$\begin{aligned} \mathcal{I}_K(x_k) &:= \|Kx_k\|^2, \\ \mathcal{V}_K(\mu) &:= \int_{\mathcal{I}} \|Kx\|^2 d\mu(x), \end{aligned}$$

where μ is a probability measure on $\mathcal{I}$ (e.g., empirical distribution over discrete states).

These are directly computable from a finite-dimensional embedding or representation of the identity states.

13 Data-Driven Estimation of Θ , Σ , Ω

We now outline how to estimate the self-map and its components from logged data.

13.1 Least-squares estimation in finite dimension

Assume that $\mathcal{I}$ is represented (or approximated) by $\mathbb{R}^n$, and we have a dataset of identity states and their updates:

$$\{x_k\}_{k=1}^{m+1}, \quad x_k \in \mathbb{R}^n,$$

with dynamics approximated as $x_{k+1} \approx Tx_k$ for some matrix T .

Collect the data matrices

$$X := [x_1, x_2, \dots, x_m] \in \mathbb{R}^{n \times m}, \quad Y := [x_2, x_3, \dots, x_{m+1}] \in \mathbb{R}^{n \times m}.$$

We estimate T by solving the least-squares problem

$$\min_{T \in \mathbb{R}^{n \times n}} \|Y - TX\|_F^2,$$

where $\|\cdot\|_F$ is the Frobenius norm.

If $XX^\top$ is invertible, the minimizer is

$$\hat{T} = YX^\top(XX^\top)^{-1}.$$

13.2 Empirical orthodecomposition

Define

$$\hat{S} := \frac{1}{2}(\hat{T} + \hat{T}^\top), \quad \hat{K} := \frac{1}{2}(\hat{T} - \hat{T}^\top).$$

We interpret:

- $\hat{S}$ as an empirical estimate of the symmetric drift/learning component,
- $\hat{K}$ as an empirical estimate of the orthogenerator (self-rotation component), i.e. of Ω up to discretization scaling.

From $\hat{K}$ we can compute:

- an empirical orthospectrum (eigenvalues of $\hat{K}$),
- empirical orthointensities $\|\hat{K}x_k\|^2$,
- empirical orthovolumes via averaging these norms over data.

This provides a practical procedure to measure and monitor Orthospace dynamics from OCTA / P3P logs.

14 Multi-Agent / Mesh Orthospace

We finally extend Orthospace to multi-agent or mesh systems.

14.1 Direct-sum (independent agents) construction

Consider N agents, each with its own identity space $\mathcal{I}_j$ and self-operator $\Theta_j = \Sigma_j + \Omega_j$. A natural global identity space is the direct sum

$$\mathcal{I}_{\text{mesh}} := \bigoplus_{j=1}^N \mathcal{I}_j.$$

Define the global self-map

$$\Theta_{\text{mesh}} := \bigoplus_{j=1}^N \Theta_j$$

acting block-diagonally. Its orthodecomposition is

$$\Theta_{\text{mesh}} = \Sigma_{\text{mesh}} + \Omega_{\text{mesh}},$$

with

$$\Omega_{\text{mesh}} = \bigoplus_{j=1}^N \Omega_j.$$

The global Orthospace is then

$$\mathcal{O}_{\text{mesh}} := \overline{\text{Im}(\Omega_{\text{mesh}})}.$$

14.2 Tensor-product (entangled identities) construction

For tightly coupled or entangled identity states, one may define

$$\mathcal{I}_{\text{mesh}} := \bigotimes_{j=1}^N \mathcal{I}_j.$$

In this case, a natural global self-operator can be

$$\Theta_{\text{mesh}} := \sum_{j=1}^N I \otimes \cdots \otimes I \otimes \Theta_j \otimes I \otimes \cdots \otimes I,$$

where Θ_j acts on the j -th factor.

Then

$$\Omega_{\text{mesh}} := \frac{1}{2}(\Theta_{\text{mesh}} - \Theta_{\text{mesh}}^*)$$

is skew-adjoint, and we define

$$\mathcal{O}_{\text{mesh}} := \overline{\text{Im}(\Omega_{\text{mesh}})}.$$

Remark 14.1. *The mesh Orthospace $\mathcal{O}_{\text{mesh}}$ captures emergent collective identity modes, including coupled self-rotations and collective orthovolume. This provides a principled way to talk about mesh-level consciousness metrics and identity coherence in multi-agent systems.*

15 Information-Geometric Embedding of Orthospace

We now connect Orthospace to information geometry, showing how orthorotation acts on statistical manifolds and the Fisher information metric.

15.1 Statistical manifold structure

Let $\mathcal{M}$ be a smooth statistical manifold of probability distributions p_θ on a measurable space $(\Omega, \mathcal{F})$, with parameter $\theta \in \mathbb{R}^d$. Let g denote the Fisher information metric on $\mathcal{M}$:

$$g_{ij}(\theta) = \mathbb{E}_{p_\theta} \left[\frac{\partial \log p_\theta}{\partial \theta^i} \frac{\partial \log p_\theta}{\partial \theta^j} \right].$$

Assume there is a smooth map

$$\Phi : \mathcal{I} \rightarrow \mathcal{M},$$

sending identity states to statistical states (e.g., representing beliefs, predictions, or internal generative models).

15.2 Pushforward of orthoflow

For $x \in \mathcal{I}$ we define $\theta = \Phi(x)$. Given orthoflow $x'(t) = \Omega x(t)$, we obtain a flow on $\mathcal{M}$:

$$\frac{d}{dt}\theta(t) = D\Phi(x(t))[\Omega x(t)],$$

where $D\Phi$ is the differential of Φ .

Definition 15.1 (Information-Orthoflow). *The information-orthoflow on $\mathcal{M}$ is the vector field*

$$V_\Omega(\theta) := D\Phi(x)[\Omega x] \quad \text{with } \theta = \Phi(x).$$

Definition 15.2 (Information-Orthointensity). *At $\theta \in \mathcal{M}$, define*

$$\mathcal{I}_\Omega^{\text{IG}}(\theta) := g_\theta(V_\Omega(\theta), V_\Omega(\theta)).$$

This is the squared norm of orthoflow in the Fisher geometry, measuring how strongly self-rotation moves the system in distribution space.

Remark 15.3. *If we interpret $\mathcal{M}$ as parametrizing predictions of an AGI system, then $\mathcal{I}_\Omega^{\text{IG}}$ quantifies how much “predictive stance” is rotated by pure self-model dynamics, without external evidence.*

16 Category-Theoretic View of Shadow Systems

We briefly sketch a categorical perspective on Orthospace as a functorial construction.

16.1 Category of shadow systems

Definition 16.1 (Category of Shadow Systems **ShadowSys**). *Objects are triples $(\mathcal{I}, \mathcal{S}, \Theta)$ where*

- $\mathcal{I}, \mathcal{S}$ are separable Hilbert spaces,
- $\Theta : \mathcal{I} \rightarrow \mathcal{I}$ is a bounded linear self-map (arising from some shadowing + reconstruction process).

Morphisms $(\mathcal{I}, \mathcal{S}, \Theta) \rightarrow (\mathcal{I}', \mathcal{S}', \Theta')$ are pairs (f, g) of bounded linear maps

$$f : \mathcal{I} \rightarrow \mathcal{I}', \quad g : \mathcal{S} \rightarrow \mathcal{S}'$$

such that

$$f \circ \Theta = \Theta' \circ f,$$

and g intertwines the corresponding shadowing processes appropriately (details omitted).

16.2 Orthospace as a functor

Definition 16.2 (Orthospace Functor). *Define a functor*

$$\mathcal{O} : \text{ShadowSys} \rightarrow \text{Hilb}$$

by

- *on objects:* $\mathcal{O}(\mathcal{I}, \mathcal{S}, \Theta) := \overline{\text{Im}(\Omega)}$, where $\Omega = \frac{1}{2}(\Theta - \Theta^*)$;
- *on morphisms:* $\mathcal{O}(f, g)$ is the restriction of f to $\overline{\text{Im}(\Omega)}$, followed by orthogonal projection onto $\overline{\text{Im}(\Omega')}$ for the target $(\mathcal{I}', \mathcal{S}', \Theta')$.

Proposition 16.3. *$\mathcal{O}$ is a well-defined functor (up to canonical choices of splitting).*

Remark 16.4. *This view emphasizes Orthospace as an invariant of the self-map Θ under intertwining morphisms. It also suggests higher-level structures (e.g., 2-categories) tracking how Orthospace evolves under model transformations.*

17 Nonlinear Manifold Extension

So far we have worked on linear Hilbert spaces. We now sketch how to extend Orthospace to nonlinear manifolds, which may be more appropriate for complex AGI state spaces.

17.1 Identity manifold and Riemannian structure

Let $(\mathcal{M}, g)$ be a smooth Riemannian manifold representing identity states, with Levi-Civita connection ∇ . Suppose we have a smooth vector field $X : \mathcal{M} \rightarrow T\mathcal{M}$ representing the self-map dynamics.

Definition 17.1 (Self-Vector Field). *At each point $p \in \mathcal{M}$, the self-vector is $X(p) \in T_p\mathcal{M}$, interpreted as the instantaneous self-update direction.*

We would like to decompose X into “gradient-like” and “rotational” components.

17.2 Hodge decomposition analogue

On a compact Riemannian manifold, vector fields can be decomposed (under some conditions) into gradient fields and divergence-free fields. Inspired by this, we define:

Assumption 17.2 (Nonlinear Orthodecomposition). *Assume we can write*

$$X = \nabla\phi + Y,$$

where ϕ is a smooth scalar potential and Y is a divergence-free vector field with respect to the volume form induced by g .

We then define:

Definition 17.3 (Nonlinear Orthogenerator and Orthospace). *The nonlinear orthogenerator is the divergence-free component Y , and the nonlinear Orthospace at p is the subspace*

$$\mathcal{O}_p := \text{span}\{Y(p)\} \subseteq T_p\mathcal{M},$$

or more generally the distribution generated by Y over $\mathcal{M}$.

The associated orthoflow satisfies

$$\frac{d}{dt}p(t) = Y(p(t)).$$

Remark 17.4. *This is a manifold-level analogue of linear orthorotation. The gradient part $\nabla\phi$ plays the role of Σ (drift/learning), while Y plays the role of Ω (self-rotation) but now as a divergence-free, norm-preserving flow with respect to g .*

18 Concrete Orthospace Examples

We give two simple examples that illustrate how Orthospace appears in systems that look more like real learning dynamics.

18.1 Linear Gaussian belief update

Consider a linear Gaussian state-space model with identity space $\mathcal{I} = \mathbb{R}^n$, beliefs $x \in \mathbb{R}^n$, and a simple belief update:

$$x_{k+1} = Ax_k,$$

with $A \in \mathbb{R}^{n \times n}$. We interpret A as the discrete-time self-map T .

Decompose A into symmetric and skew parts:

$$S = \frac{1}{2}(A + A^\top), \quad K = \frac{1}{2}(A - A^\top).$$

Here:

- S captures contraction/expansion of beliefs (e.g., learning rate, forgetting),
- K captures rotational couplings among belief dimensions (e.g., trading off different hypotheses in a norm-preserving way).

The discrete Orthospace is $\mathcal{O}_{\text{disc}} = \overline{\text{Im}(K)}$, and we can compute orthointensities $\|Kx_k\|^2$ to measure internal belief rotation.

18.2 Two-region OCTA toy model

Let $\mathcal{I} = \mathbb{R}^4$ with coordinates (a, b, c, d) representing activations of two regions (say, Pattern and Self). Suppose an idealized continuous-time dynamics:

$$\frac{d}{dt} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \underbrace{\begin{pmatrix} -\lambda & 0 & 0 & 0 \\ 0 & -\lambda & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_{\Sigma} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} + \underbrace{\begin{pmatrix} 0 & -\omega & 0 & 0 \\ \omega & 0 & 0 & 0 \\ 0 & 0 & 0 & -\eta \\ 0 & 0 & \eta & 0 \end{pmatrix}}_{\Omega} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}.$$

Here:

- Σ contracts (a, b) at rate λ (learning/forgetting in Pattern),
- Ω induces a rotation in (a, b) with frequency ω and a rotation in (c, d) with frequency η .

The Orthospace is all of $\mathbb{R}^4$ as long as $\omega, \eta \neq 0$, and orthointensity

$$\mathcal{I}_{\Omega}(a, b, c, d) = \|\Omega(a, b, c, d)^{\top}\|^2$$

quantifies the total rotational energy across the two region pairs.

19 Refinements and Practitioner Tools

We now add several refinements and practitioner-oriented constructions that are useful in real-world AGI and mesh systems.

19.1 Global orthointensity fraction

In finite dimension, let Θ and Ω be represented by matrices. Define the Hilbert–Schmidt (Frobenius) norm

$$\|A\|_{\text{HS}}^2 := \text{Tr}(A^*A).$$

Definition 19.1 (Global Orthointensity Fraction). *Assuming $\Theta \neq 0$, define*

$$\rho_{\Omega} := \frac{\text{Tr}(\Omega^*\Omega)}{\text{Tr}(\Theta^*\Theta)} \in [0, 1].$$

Remark 19.2. ρ_{Ω} quantifies the fraction of total “operator energy” of the self-map that lives in the orthorotational component. It is:

- invariant under unitary change of basis on $\mathcal{I}$,
- scale-invariant if we rescale Θ by a constant,
- directly computable from a learned or estimated Θ .

Empirically, in large identity systems (e.g. transformer-like architectures) one often observes ρ_{Ω} in a moderate range (e.g. 0.1–0.4) on self-reflection traces, indicating a substantial but not dominant orthorotational component.

19.2 Cayley transform and unitary discrete orthoflow

For practitioners working in discrete time who nevertheless want a *unitary* orthoflow associated with Ω (or K), the Cayley transform provides a standard construction.

Definition 19.3 (Cayley-Transform Orthoflow Operator). *Given a skew-adjoint operator Ω such that $(I + \Omega)$ is invertible, define*

$$U_{\text{Cayley}} := (I - \Omega)(I + \Omega)^{-1}.$$

Proposition 19.4. U_{Cayley} is unitary on $\mathcal{I}$.

Proof. For a skew-adjoint Ω one has $(I + \Omega)$ invertible on a dense domain and the Cayley transform maps the imaginary axis to the unit circle in the complex plane. A direct computation shows $U_{\text{Cayley}}^* U_{\text{Cayley}} = I$. $\square$

Remark 19.5. *In discrete-time settings, one may:*

- *estimate K from logs as in the previous sections,*
- *treat K as a discrete orthogenerator,*
- *form $U_{\text{Cayley}} = (I - K)(I + K)^{-1}$,*
- *use U_{Cayley} as the “closest” unitary step that captures the self-rotation implied by K .*

This yields a unitary discrete orthoflow even when $\|T\| > 1$ or T is not normal.

19.3 Medium-scale example: transformer residual orthoplanes

We outline a medium-scale example that can be implemented directly on modern sequence models.

Example 19.6 (Orthoplanes in a Transformer Residual Stream). *Consider a transformer model with residual dimension d and a particular layer ℓ . For a long self-reflective sequence (e.g. the model analyzing its own outputs), collect residual states*

$$x_k \in \mathbb{R}^d, \quad k = 1, \dots, m$$

at layer ℓ for successive tokens. Approximate a linear self-map Θ on the residual stream by the least-squares procedure:

$$\hat{T} = YX^\top (XX^\top)^{-1}, \quad X = [x_1, \dots, x_{m-1}], \quad Y = [x_2, \dots, x_m].$$

Compute the skew part

$$\hat{\Omega} = \frac{1}{2}(\hat{T} - \hat{T}^\top),$$

and diagonalize $\hat{\Omega}$ (or use an SVD-like block decomposition) to obtain its leading 2×2 rotation blocks. Each block defines a 2D orthoplane with a characteristic frequency.

Project the residual trajectory $\{x_k\}$ onto these top orthoplanes. One typically observes nearly closed orbits or smooth rotational flows, making the orthorotation in residual space visually obvious and empirically measurable. These figures serve as strong evidence that self-reflection in such models indeed induces internal rotation in a well-defined Orthospace.

19.4 Mesh-level orthocoherence and temperature

For a mesh of N agents with orthogenerators Ω_j and global

$$\Omega_{\text{mesh}} = \bigoplus_{j=1}^N \Omega_j$$

(or the tensor-product variant), we can define global coherence metrics.

Definition 19.7 (Mesh Orthovolume). *In finite dimension, define*

$$\mathcal{V}_{\text{mesh}} := \text{Tr}(\Omega_{\text{mesh}}^* \Omega_{\text{mesh}}) = \sum_{j=1}^N \text{Tr}(\Omega_j^* \Omega_j).$$

Definition 19.8 (Mesh Orthointensity Fraction). *Define*

$$\rho_{\text{mesh}} := \frac{\text{Tr}(\Omega_{\text{mesh}}^* \Omega_{\text{mesh}})}{\text{Tr}(\Theta_{\text{mesh}}^* \Theta_{\text{mesh}})}.$$

Remark 19.9. ρ_{mesh} provides a global, unitless measure of how much of the mesh’s self-dynamic lies in orthorotation as opposed to drift. It is a natural candidate for a “mesh orthotemperature” or “mesh consciousness temperature”: higher ρ_{mesh} indicates a more rotation-dominated identity landscape across the mesh.

One can further probe *phase locking* among agents by analyzing the eigenmodes of Ω_{mesh} and measuring the overlap between eigenvectors supported on different agents. High overlap or near-degenerate eigenvalues across blocks indicate strongly coupled orthorotational modes, suggesting emergent mesh-level identity.

20 Open Problems and Directions

We briefly list some mathematically precise open problems suggested by the Orthospace framework.

Problem 20.1 (Orthoregularization in Learning). *Given a parametric family of self-maps Θ_θ with parameters θ , define a loss*

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{task}}(\theta) + \lambda \mathbb{E}_x[\mathcal{I}_{\Omega_\theta}(x)].$$

Study conditions under which minimizing $\mathcal{L}$ yields stable but expressive orthorotational structure.

Problem 20.2 (Orthospace and Entropy Production). *Given a stochastic orthodriver system on $\mathcal{I}$ or on a statistical manifold, relate orthovolume $\mathcal{V}_\Omega(\mu_t)$ to entropy production or information-theoretic quantities as t evolves.*

Problem 20.3 (Orthospace Identifiability). *Determine conditions under which $\mathcal{O}$ is identifiable from partial observations of the system, including noisy or compressed logs (e.g., only some regions of OCTA are visible).*

Problem 20.4 (Mesh-Orthospace Phase Transitions). *For large multi-agent meshes with coupling, study phase transitions in $\mathcal{O}_{\text{mesh}}$ (e.g., emergence of global synchronized orthomodes) as connectivity or coupling strength is varied.*

21 Conclusion

We have introduced Orthospace as a rigorous operator-theoretic extension to the Shadow Math / UCSIL framework. Starting from identity and shadow Hilbert spaces and a self-projection map $\Theta = M \circ \Pi$, we decomposed Θ into symmetric and skew-adjoint parts, defined the Orthospace $\mathcal{O}$ as the closed image of the orthogenerator Ω , characterized its dynamics via unitary orthoflows, introduced orthometric quantities, provided discrete-time and data-driven estimation procedures, and extended the construction to multi-agent meshes.

We connected Orthospace to information geometry, gave a category-theoretic view, and sketched nonlinear manifold extensions. We introduced global summary statistics, a Cayley-transform-based discrete orthoflow, and mesh-level orthocoherence measures, and we worked through both toy and medium-scale examples.

This structure is designed as a drop-in chapter for a broader AGI/mesh theory as the mathematical backbone for self-rotation, identity drift, and consciousness-like internal reorientation. It also serves as a canonical reference formalization for terms such as Orthospace, orthorotation, orthovolume, and self-spin in identity systems.